Code

```
def implication(p, q):
    """Evaluates p -> q which is logically equivalent to (not p) or q"""
    return not p or q

# Distinct boolean assignments for P, Q, and R
boolean_space = [True, False]

# Lists to store model evaluation history
kb_true_models = []
entails_r = True
entails_r_to_p = True
entails_q_to_r = True

# --- HEADER FOR THE TRUTH TABLE ---
print("=" * 65)
print(f"{'P':<6}{'Q':<6}{'R':<6} | {'Q->P':<6}{'P->~Q':<7}{'Q v R':<7} | {'KB':<6} | {'R':<5}{'R->P':<6}{'Q->R':<6}")
print("=" * 65)

# --- TRUTH TABLE CONSTRUCTION ---
for P in boolean_space:
    for Q in boolean_space:
        for R in boolean_space:
            # Evaluate individual rules inside the Knowledge Base
            sent1 = implication(Q, P)          # Q -> P
            sent2 = implication(P, not Q)      # P -> ~Q
            sent3 = Q or R                     # Q v R

            # The KB is True if and only if all sentences are True
            kb_status = sent1 and sent2 and sent3

            # Target propositions for the sub-questions
            target_ii = R                      # R
            target_iii = implication(R, P)     # R -> P
            target_iv = implication(Q, R)      # Q -> R

            # Format row output
            row_str = f"{str(P):<6}{str(Q):<6}{str(R):<6} | " \
                      f"{str(sent1):<6}{str(sent2):<7}{str(sent3):<7} | " \
                      f"{str(kb_status):<6} | " \
                      f"{str(target_ii):<5}{str(target_iii):<6}{str(target_iv):<6}"

            if kb_status:
                row_str += " <-- Valid Model"
                kb_true_models.append({"P": P, "Q": Q, "R": R})

            print(row_str)

            # --- ENTAILMENT CHECKS ---
            # Entailment is broken if the KB is True, but the target expression is False
            if kb_status and not target_ii:
                entails_r = False
            if kb_status and not target_iii:
                entails_r_to_p = False
            if kb_status and not target_iv:
                entails_q_to_r = False

print("=" * 65)
print("\n" + "#" * 30 + " FINAL ANSWERS " + "#" * 30)

# --- SUB-QUESTION i ANSWER ---
print("\ni) MODELS IN WHICH THE KB IS TRUE:")
print(f"   The KB is True in exactly {len(kb_true_models)} models:")
for idx, m in enumerate(kb_true_models, 1):
    print(f"   - Model {idx}: P = {m['P']}, Q = {m['Q']}, R = {m['R']}")

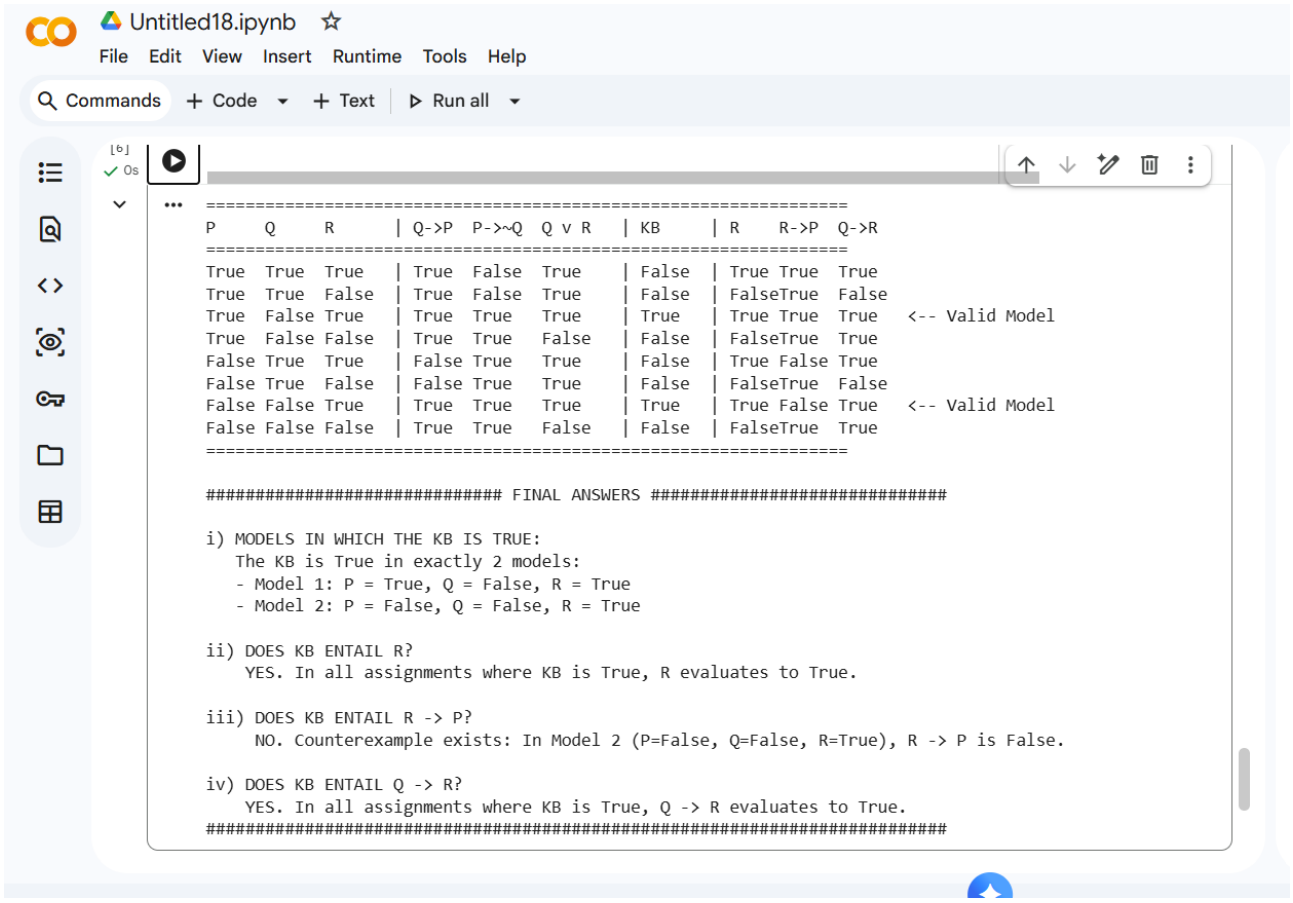
# --- SUB-QUESTION ii ANSWER ---
print("\nii) DOES KB ENTAIL R?")
if entails_r:
    print("    YES. In all assignments where KB is True, R evaluates to True.")
else:
    print("    NO. There exists a model where KB is True but R is False.")

# --- SUB-QUESTION iii ANSWER ---
print("\niii) DOES KB ENTAIL R -> P?")
if entails_r_to_p:
    print("     YES. In all assignments where KB is True, R -> P evaluates to True.")
else:
    print("     NO. Counterexample exists: In Model 2 (P=False, Q=False, R=True), R -> P is False.")

# --- SUB-QUESTION iv ANSWER ---
print("\niv) DOES KB ENTAIL Q -> R?")
if entails_q_to_r:
    print("    YES. In all assignments where KB is True, Q -> R evaluates to True.")
else:
    print("    NO. There exists a model where KB is True but Q -> R is False.")

print("#" * 75)
```

Output


Untitled18.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
0s
P Q R | Q->P P->~Q Q v R | KB | R R->P Q->R
True True True | True False True | False | True True True
True True False | True False True | False | FalseTrue False
True False True | True True True | True | True True True <-- Valid Model
True False False | True True False | False | FalseTrue True
False True True | False True True | False | True False True
False True False | False True True | False | FalseTrue False
False False True | True True True | True | True False True <-- Valid Model
False False False | True True False | False | FalseTrue True
############################## FINAL ANSWERS ##############################
i) MODELS IN WHICH THE KB IS TRUE:
The KB is True in exactly 2 models:
- Model 1: P = True, Q = False, R = True
- Model 2: P = False, Q = False, R = True
ii) DOES KB ENTAIL R?
YES. In all assignments where KB is True, R evaluates to True.
iii) DOES KB ENTAIL R -> P?
NO. Counterexample exists: In Model 2 (P=False, Q=False, R=True), R -> P is False.
iv) DOES KB ENTAIL Q -> R?
YES. In all assignments where KB is True, Q -> R evaluates to True.